Emergent Complexity – An investigation

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1 Introduction and Investigated Questions

Local update rules can generate cycles of patterns. Programmed, controlled automaton provide a controlled setting for studying this relationship. The questions asked involves the categories of the final results and its size, the addition of intermediate states, the introduction of randomness and its effect on the end result. With that said, this report does not attempt to generalize this to a universal ruling or pattern, I found much beauty in the chaos of most of the seeds.

2 Mathematical Background

Let $x_t(i) \in \{0, \ldots, C-1\}$ be the state of cell i on a $W \times H$ grid. And the set $N(i)$ be the set of neighboring cells of cell i. The eight neighbors wrap around the grid, which is equivalent to modulo the height and width. The active neighbor count is

$$n_t(i) = \sum_{j \in N(i)} [x_t(j) = 1].$$

In binary mode the only states are dead and alive. A separate value stores age, capped at 255 to avoid memory overhead; a cell that is alive but "aging" are ignored in the calculations of $n_t(i)$ and checking whether repeated states, more specifically, an aging cell is counted as dead in calculating scenarios. A binary rule consists of two independent 9-bit masks, giving $2^{18} = 262144$ possible rules. Encoding is $m_B = \sum_{n \in B} 2^n$ and $m_S = \sum_{n \in S} 2^n$. This includes B0 rules, which albeit not useful, are cyclic in nature.

3 System Architecture

This section is largely done with Codex. This is largely due to my lack of familiarity with website construction. My prompt asked for a website infrastructure that implements most of the website architecture without implementing the main engine for the automaton. The main engine is implemented in cpp by myself.

`cpp/engine.cpp` keeps tracks of parsing, initialization, seeded randomness and state transitions. `cpp/experiments.cpp` keeps tracks of measurements, state tracking (repetition) and categories of the end result. Embind exposes copies of typed cell buffers and JSON trials. JavaScript handles the canvas, controls, worker scheduling and downloads. A Web Worker keeps the interface alive while a trial runs; cancellation terminates the worker and preserves completed trials.

The following paragraph is pasted from Codex:

CMake and Emscripten build a static eight-file site in `build/wasm/site`. HTML assets use relative URLs; the worker and Wasm loader resolve siblings relative to their modules. HTTP testing covers both root hosting and a repository subpath. The GitHub Actions workflow builds from C++ and uploads the staged directory to Pages. The Node preview server is development infrastructure, not a deployed backend. The public GitHub-hosted deployment itself requires repository settings and a push; local tests cannot establish its success.

4 Rule Implementation

As mentioned in Section 2, each rule can be represented as a 9-bit bitstring. Take some examples, B3/S23 is Conway Life; B36/S23 is HighLife. There are no restrictions on the ruleset. For example, B/S is valid for representing an empty grid. Some sanity checking was done in the main parser: duplicate digits, in range, correct format. Checkboxes and a text entry edit the same masks.

A binary wrapper was implemented to speed up state transitioning in non-binary modes, this wrapper implements a feature mentioned in Section 2: it defaults all aging cells to dead but occupied in calculations. This retains the original display implemented in Part A and exact occupancy behavior. Unit tests have coverage over every combination for all 18 individual mask bits.

5 Transitioning with Multiple States

For $C \in \{3, \ldots, 8\}$, 0 is inactive, 1 active, and 2 through $C-1$ dead but occupying space. Only state 1 counts as a live neighbor. Let us establish that, for a cell with x with n neighbors, and the ruleset of birth be B, and ruleset of survival be S. The formal rule for update and propagation is

$$F(x,n) = \begin{cases} 1 & x = 0 \text{ and } n \in B, \\ 0 & x = 0 \text{ and } n \notin B, \\ 1 & x = 1 \text{ and } n \in S, \\ 2 & x = 1 \text{ and } n \notin S, \\ x+1 & 2 \leq x < C-1, \\ 0 & x = C-1. \end{cases}$$

At this point, another constant C be introduced, denoting the number of generations. Thus B3/S23/C5 specifies five states of all cells. Brian's Brain, B2/S/C3, has no active survival: an active cell always becomes aging at the next state. Refractory Life, B3/S23/C5, was added as a designed comparison because survival can maintain active cells while a failed survivor prevents immediate rebirth for three steps as it effectively dead. Different age durations can change spatial interactions.

The mass production of the experiments to generate more results have more nuances with UI handling. The safe range for C is $[2, 8]$, and dimensions limit to $[3, 128]$. Initialization with only active state assigns state 1 with probability d and 0 otherwise. Initialization with all states assigns a nonzero state with probability d, then distributes it approximately uniformly over 1 through $C-1$ using a random number generator. Consequently, with this type of initialization, expected initial active density is $d/(C-1)$ rather than d. Comparing these initializers at equal d controls occupied density instead of active density.

6 Experimental Methodology

All main experiments use a 64 by 64 wrap-around grid, 500 updates, and nonzero-density parameters of 0.10, 0.30 and 0.50. Random sampling uses master seed 2026. Each sampled rule has five trials per density. Follow-up binary experiments use ten trials per density and noise level. Multi-state experiments cross B2/S and B3/S23, C=3,5,8, both initializers, three densities, five trials, and five noise levels.

For trial index j starting at zero, the initial seed is

$$s(d, j) = \big(2026 + (j + 1)2654435761 + \text{round}(10^6 d)2246822519\big) \bmod 2^{32}.$$

The rule index is deliberately excluded from this formula to implement controlled initial conditions, and is recorded separately. Noise starts from s xor `0x9e3779b9`. The generator is the explicit 32-bit recurrence $z' = 1664525z + 1013904223 \pmod{2^{32}}$. Most numbers used in the generation algorithms are standard for linear congruential generators. Binary events compare its output against $\lfloor p2^{32} \rfloor$. Seeds yield independent sampling in the engineering sense, there is no gurantee of generalization after conclusion. LCG quality limits the design.

The algorithm is as following:

1. Sample rules and save the rule list.
2. Derive the paired initial seeds for each density and trial.
3. Advance each configuration for a long runtime while recording measurements.
4. Classify each trajectory using predetermined categories.
5. Aggregate only sampled-rule trials for rankings.
6. Rerun the top candidate with additional seeds and noise.
7. Export configurations, grids, and summaries.

Every generation, including generation zero, records active and occupied density, each state population, births, deaths, semantic change fraction, active area divided by grid area: the smallest axis-parallel box containing every cell, coordinate center-of-mass displacement divided by the grid diagonal producing a metric approximating velocity of the movement of the cells, and a grid hash. Births/deaths count changes of active status after noise; they include perturbation effects. The initial transition measurements are zero.

Let $w = \max(1, \lfloor T/4 \rfloor)$, where T denotes total generation of 500 denoted earlier in this section. Growth is the difference between mean density at $T - w + 1, \ldots, T$ and $0, \ldots, w - 1$. Dividing by $T - w + 1$ gives a rate per generation between window centers. Spatial growth applies the same calculation to bounding-box area. Signal of stagnant activity are the corresponding window mean change percentage and mean centroid movement. Final, mean and maximum active density are also exported. No early stopping is used.

7 Rule Sampling and Classification

The sampler takes the high 18 bits of successive LCG draws, splits the bottom nine bits into B and the top nine into S, and rejects previously seen rules until exactly 100 distinct rules have been collected. This is a reproducible pseudo-random sample without replacement. Life and HighLife are run separately. Controls do not enter sample statistics or rankings.

Classification priority is deterministic and exported in the C++ Criteria configuration: maxPeriod=16, growth=0.10, disorder=0.15, persistent=0.001. In order:

1. Extinct (export label `extinction`): final active count zero, p=0 and B0 absent. Refractory states may remain briefly, but cannot create future active cells.
2. Fixed point (`stable`): terminal exact period 1 at p=0.
3. Periodic (`oscillatory`): terminal exact period 2–16 at p=0.
4. Expanding (`growing`): both density and spatial growth exceed 0.10.
5. Highly active (`disordered`): last-window change fraction exceeds 0.15.
6. Bounded persistent activity: final activity is nonzero and last-window change exceeds 0.001 after the earlier tests fail.
7. Unclassified: all other cases; the horizon is reached without classification.

"Bounded" refers only to this finite observation and grid, not a bound on an infinite-plane trajectory. Dense random initialization already spans most of the grid, making the expansion category conservative. A rule can have several trial labels, and its summary is only a dominant observed category.

Hashes accelerate recurrence search over 16 prior states processed in batch. A hash match always requires full-grid equality. The exported first recurrence generation describes the earliest observed match; terminalPeriod describes the match at generation 500. These need not be the same event. Grid recurrence under noise is an observation, not proof of a periodic trajectory, because RNG state is not included in that comparison. The firstEmpty field records any first active-empty generation, even when B0 or noise can later reactivate the grid.

7.1 Formulation of Scores and Justification

To systematically rank rules exhibiting structured, non-trivial dynamics, each trial is scored using an empirical heuristic I, formulated as

$$I = \mathbf{1}(0 < \rho_T < 0.9) + \mathbf{1}(P_T > 2) + \max(0, G_A) + \min(1, 20M) + 4a(1-a), \tag{1}$$

supplemented by an aggregate rule-level multistability bonus of $(K-1)/6$ added to the sample mean $\bar{I}$. The constituent terms target specific dynamical regimes:

- **Viable Population Density** ($\mathbf{1}(0 < \rho_T < 0.9)$): An indicator function filtering out trivial terminal attractors. It returns 1 if the terminal active density ρ_T lies strictly within $(0, 0.9)$, penalizing both total extinction ($\rho_T = 0$) and lattice oversaturation ($\rho_T \geq 0.9$).
- **Higher-Order Periodicity** ($\mathbf{1}(P_T > 2)$): Rewards complex cyclic attractors with terminal periods $P_T \in [3, 16]$. By demanding $P_T > 2$, it filters out stationary still-lifes ($P_T = 1$) and trivial period-2 oscillators (e.g., standard blinkers).
- **Spatial Expansion** ($\max(0, G_A)$): Measures the net rate of outward colonization per generation, where G_A is the normalized difference in active bounding-box area between the initial and terminal temporal windows. The $\max(0, \cdot)$ operator guarantees that shrinking structures are not penalized.

- **Translational Mobility** ($\min(1, 20M)$): Detects propagating localized structures (such as gliders and spaceships). Because the mean coordinate center-of-mass displacement M across the grid diagonal is typically subtle ($M \ll 0.1$), the scalar multiplier 20 maps realistic velocities to an order-one reward, while the upper bound of 1.0 caps artificial spikes induced by wrap-around movements.
- **Critical Activity Turnover** ($4a(1-a)$): A symmetric logistic parabola evaluated on the final-window change fraction $a \in [0, 1]$. Maximized at $a = 0.5$, this term assigns zero reward to completely frozen lattices ($a \to 0$) and uncorrelated high-entropy churn ($a \to 1$), favoring intermediate states capable of signal transmission and potential to possess potential complexity.
- **Diversity Bonus** ($\frac{K-1}{6}$): An ensemble-level correction evaluated across all seeds and initial densities for a given rule. Here, $K \in \{1, \ldots, 7\}$ denotes the count of unique behavioral classifications observed. Because there are seven mutually exclusive classes, dividing by 6 maps the bonus strictly to the interval $[0, 1]$, favoring rules with sensitive, initial-condition-dependent phase spaces over monolithic dynamics.

8 Results

The completed v2 campaign contains 2880 trials: 1500 random-sample trials, 30 separate controls, 450 binary follow-up/noise trials, and 900 Generations trials. All used master seed 2026 and reached 500 updates.

Category	Trials	Percent
disordered	1051	70.07
oscillatory	215	14.33
stable	192	12.80
bounded persistent activity	42	2.80

Table 1: Random sample only; 100 rules times three densities times five trials. Extinction, growth and unclassified counts were zero. Labels follow the stated priority.

No sampled trial was classified extinct. This sample-specific absence does not establish persistence for other seeds, sizes or rules. 24 sampled rules had more than one observed category. Spatial expansion was not detected by the conservative bounding-box criterion; random initial grids already occupied most of the available spatial extent.

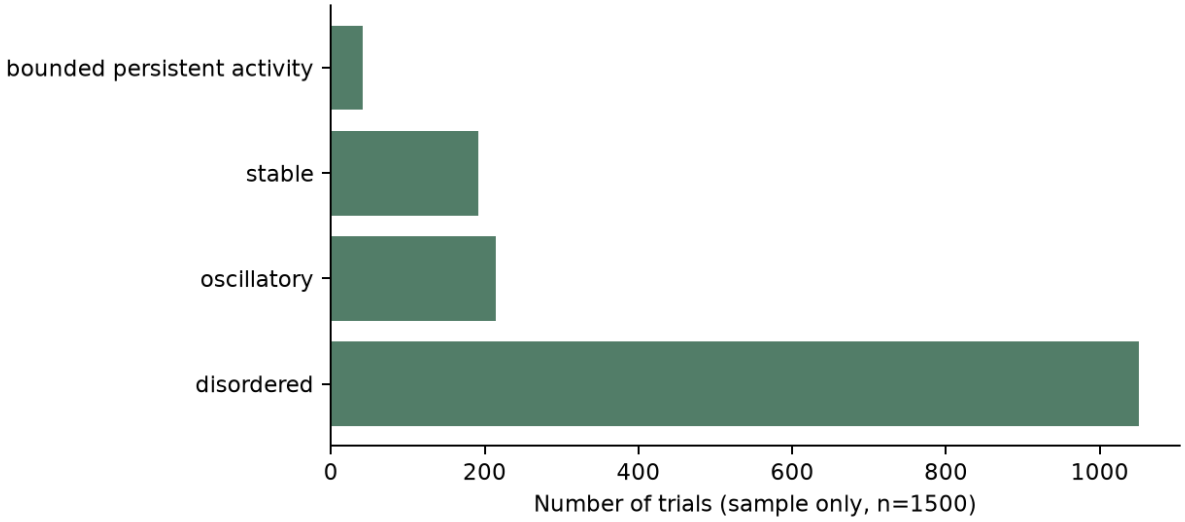


Figure 1: Measured classification distribution. Categories describe trajectories, not intrinsic rule classes.

Rule	C	Initializer	Active	Nonzero	Extinct / 15
B2/S	3	active-only	0.0176	0.0349	0
B2/S	3	all-states	0.0147	0.0296	0
B2/S	5	active-only	0.0015	0.0061	8
B2/S	5	all-states	0.0026	0.0102	1
B2/S	8	active-only	0.0008	0.0059	10
B2/S	8	all-states	0.0005	0.0036	7
B3/S23	3	active-only	0.0233	0.0233	0
B3/S23	3	all-states	0.0196	0.0196	1
B3/S23	5	active-only	0.0196	0.0196	0
B3/S23	5	all-states	0.0063	0.0063	4
B3/S23	8	active-only	0.0213	0.0223	0
B3/S23	8	all-states	0.0019	0.0019	8

Table 2: Zero-noise Generations outcomes pooled equally across the three densities (five trials per density). Full per-density outcomes remain in the JSON.

The active and total nonzero columns separate activity from refractory occupancy. Differences between C values and B/S rules concern state dynamics. Initialization changes expected starting active density.

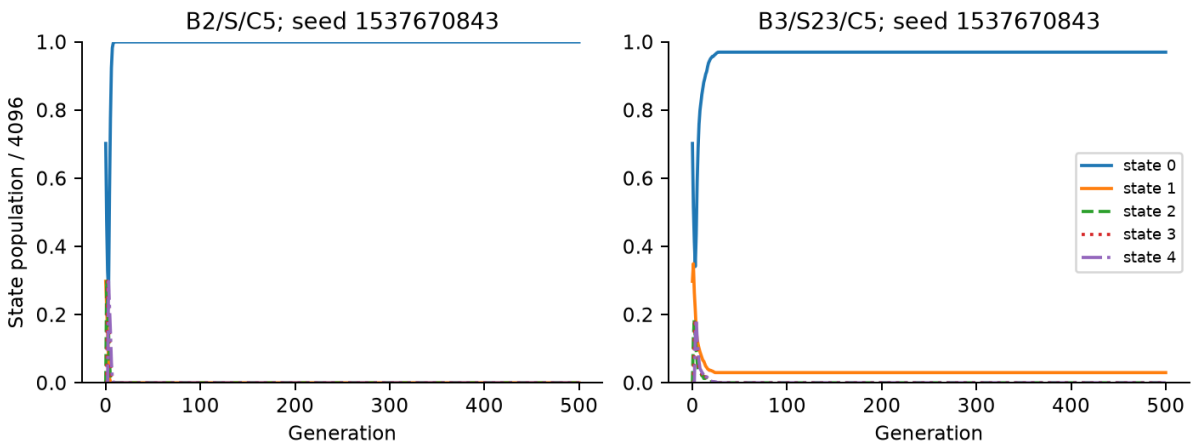


Figure 2: Measured per-state density histories for trial 0, nonzero density 0.30, active-only initialization and zero noise. Dead state 0 is included; these traces distinguish active persistence from refractory decay.

For zero-noise B2/S across C and initializers, the measured categories were 64 bounded persistent activity, 26 extinction.

For zero-noise B3/S23 across C and initializers, the measured categories were 70 stable, 13 extinction, 3 oscillatory, 4 bounded persistent activity.

Exact fixed points and recurrences support grid-level stable or cyclic structure claims. Againg population changes alone are not counted as emergent structure. Table 1 and Figure 1 summarize the random sample. Table 2 separates active and refractory occupancy.

9 Detailed investigation of selected rules

The highest-ranked sampled candidate was B078/S03568, with interest score 2.3544. Its original 15 sample trials had the distribution 12 oscillatory, 3 bounded persistent activity. It was selected for mixed observed behavior and nontrivial recurrence, not biological resemblance. B0 also allows spontaneous activation from empty regions.

Density	Mean final	Final range	Observed categories
0.1	0.2271	0.2192–0.2351	7 oscillatory; 3 persistent
0.3	0.2343	0.2249–0.2424	9 oscillatory; 1 persistent
0.5	0.2355	0.2236–0.2515	7 oscillatory; 3 persistent

Table 3: Selected-rule zero-noise follow-up: ten trials per density. The category distribution reports both common and exceptional outcomes.

The additional trials give 23 oscillatory, 7 bounded persistent activity. Terminal periods across these runs were 0, 6, 12. Period zero means no recurrence in the 16-step terminal window. Exact repeating grids support a cyclic structure claim for the deterministic recurrent trials; no translating object or reproduction claim follows. Table 3 reports density-specific follow-up variation and

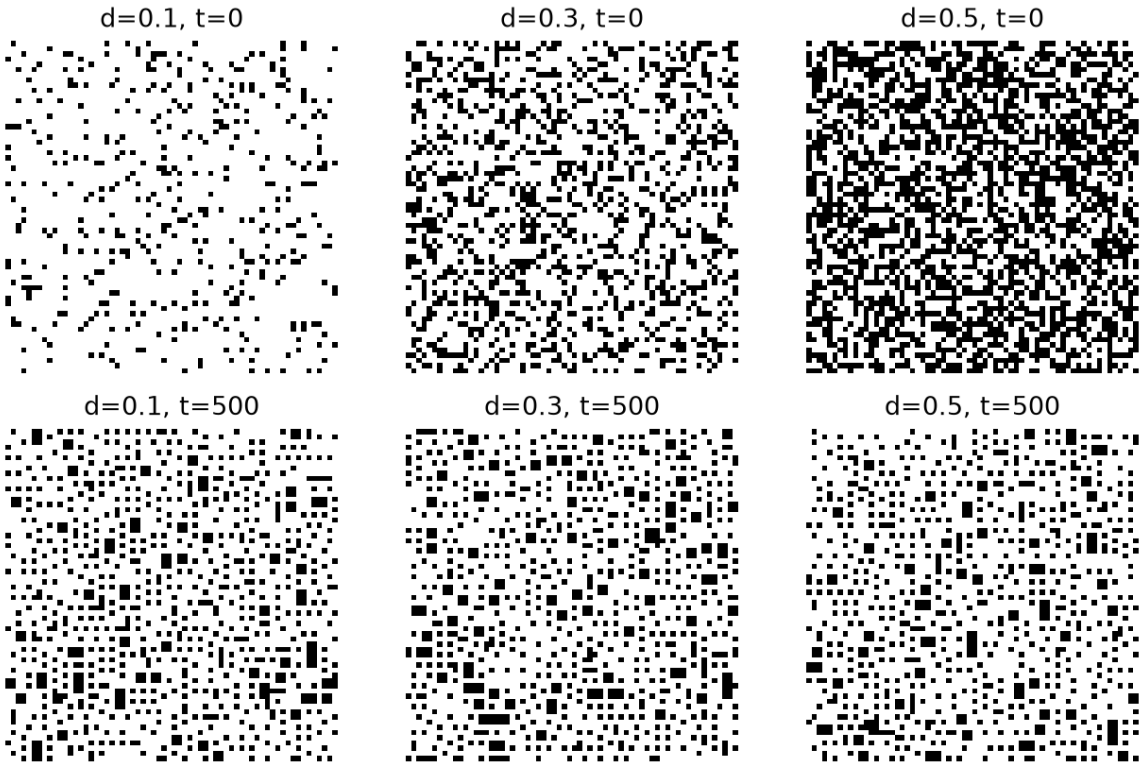


Figure 3: Actual B078/S03568 grids, trial 0 at each density, p=0. Top: initialization; bottom: generation 500. Exact derived seeds are in the replay files. This figure is generated with the aid of Codex

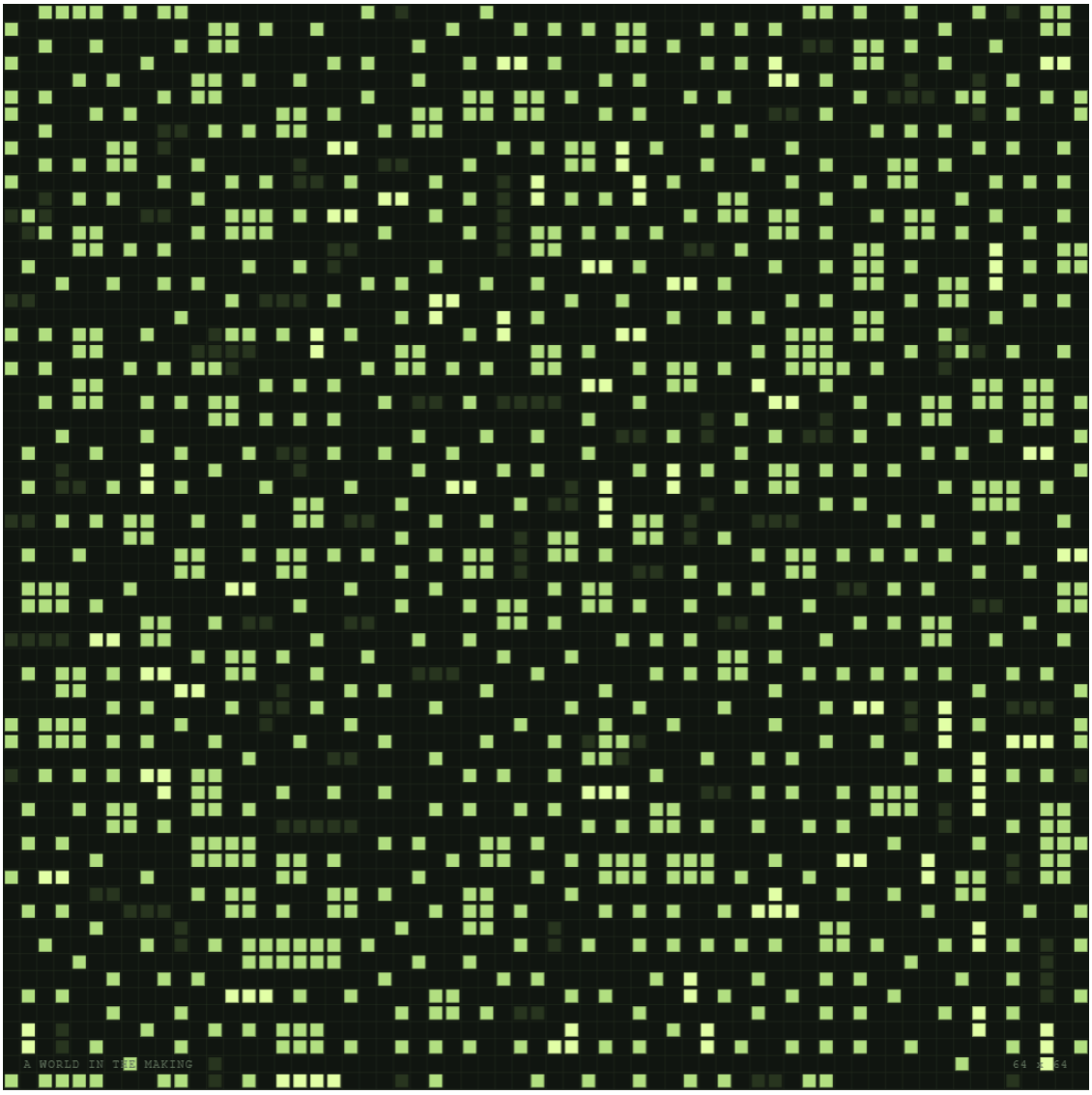


Figure 4: Working application screenshot after 100 exact replay steps of the selected sampled rule, density 0.30, trial 0, zero noise.

Figure 3 shows actual saved grids. The higher-trial follow-up retains the original three densities and extends paired trials from five to ten. Typical and exceptional labels are both retained in the generated tables and raw data. Exact replay JSON files preserve three examples at each density for zero noise. These can be reopened in the application. The screenshots and snapshots accompanying this report are generated from those actual grids. Pattern persistence is a grid-level observation; no parent-offspring or object correspondence is inferred from it.

10 Noise Investigation

For each cell, first compute $F(x, n)$ synchronously. With probability p, binary mode complements the result. In Generations mode, replace the deterministic result uniformly with one of the other C-1 states. Rejection sampling removes modulo bias from this replacement. All cells use data from the previous generation; perturbations are applied to the new generation, so they influence neighbors only in the next update. At p=0 no RNG values are consumed and the deterministic result is preserved exactly.

The levels are $0, 10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}$. A 64 by 64 grid therefore expects 0, 0.04096, 0.4096, 4.096 or 40.96 perturbations per update. Each noisy run is also compared to a separately advanced deterministic trajectory with the same initial grid. Full-state disagreement is measured in all sample states, not just active disagreement (ie dead or alive). Because different probabilities stem from different numbers of draws, noise histories at different p values are paired by initial seed, not identical perturbation locations.

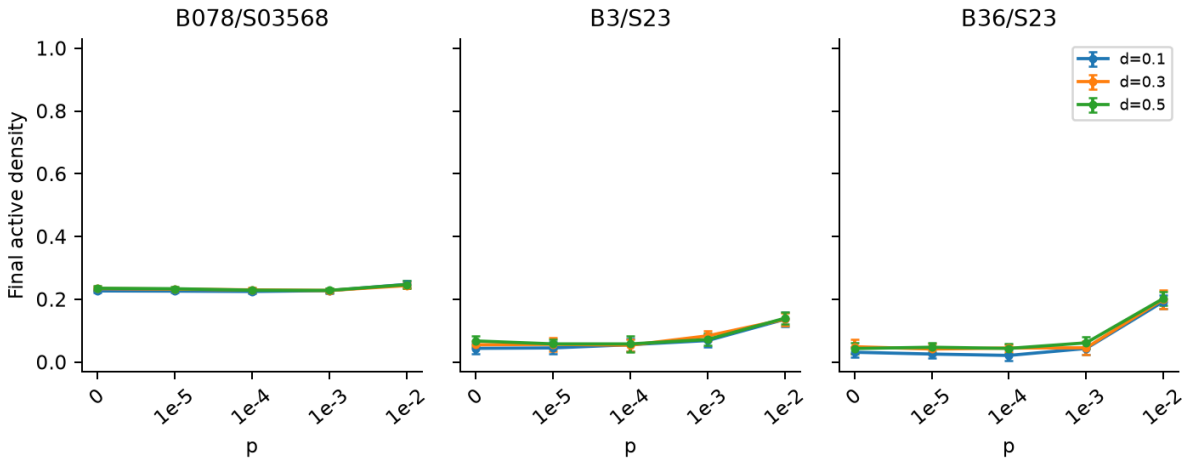


Figure 5: Binary noise response. Points are means of ten trials per density and p; error bars are one sample standard deviation. Discrete levels are equally spaced for legibility.

p	Mean density	SD	Distance	Active / 30
0	0.2323	0.0076	0.0000	30
1e-05	0.2310	0.0066	0.0210	30
0.0001	0.2283	0.0072	0.1457	30
0.001	0.2283	0.0073	0.3507	30
0.01	0.2463	0.0110	0.3646	30

Table 4: Selected-rule noise outcomes pooled across densities. SD includes between-density variation; the plot separates densities. Distance is final disagreement from the matched deterministic grid.

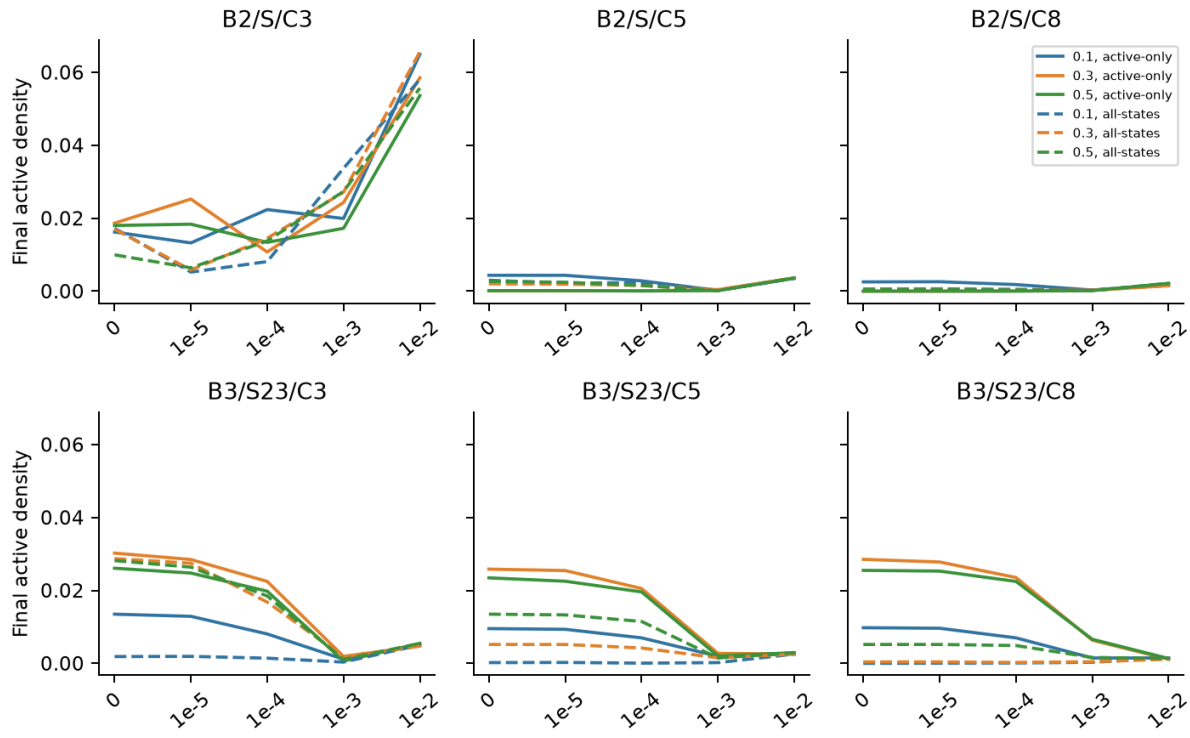


Figure 6: Generations response: means of five trials per point. Colors denote initial nonzero density; solid lines use active-only initialization and dashed lines all-states. Raw trial variation and per-state populations are exported. Lines guide comparison and do not locate a critical probability. This figure is made with Codex

For B2/S pooled across C, initialization and density, the mean final active density was 0.0063 at p=0 and 0.0216 at p=0.01. Zero-noise active extinction occurred in 26 of 90 runs.

For B3/S23 pooled across C, initialization and density, the mean final active density was 0.0153 at p=0 and 0.0030 at p=0.01. Zero-noise active extinction occurred in 13 of 90 runs. Figures 5 and 6 show the measured noise responses; Table 4 gives selected-rule numerical summaries.

The repeated outcomes and their variation are the evidence for further detailed research. Visible recovery under noise may reflect external activation rather than autonomous maintenance.

11 Tangent on Philosophy and Formal Verification

Finite local transition properties permit exhaustive testing: for each center state and each active-neighbor count, the specified next state can be compared to the implementation. Tests already exhaust the 18 binary mask-bit choices and use small fixtures for aging advancement, Brian's Brain and known Life patterns. Shape and range validation prevent out-of-range explicit states. The invariant is that each update preserves grid length and keeps semantic values in 0 through C-1. Population counts sum to the number of cells. These tests are executable evidence, not formal verification of the entire simulator or its compiler.

Deterministic replay checks that seed, initial grid and parameters reproduce an identical trajectory. Bounded-grid model checking could enumerate all states of a tiny grid and verify safety invariants or reachability, ideally using an independently generated transition relation. There are C^{WH} semantic grids. At zero noise, deterministic evolution on a finite grid must eventually recur by the pigeonhole principle; this establishes eventual periodicity, not a useful bound on cycle length.

State-space explosion prevents exhaustive checking at the experimental sizes. Computation-capable cellular automata on unbounded domains also admit undecidable long-term questions; this does not imply that a specified finite-grid trace is undecidable. The distinction between bounded implementation correctness and general emergent behavior is essential.

12 Self-replication and ecological complexity

A repeating pattern returns to the same configuration; a translating pattern recurs after a spatial shift; pattern growth increases occupied area or material. Template copying duplicates an arrangement through some causal mechanism. Genuine self-reproduction requires identifiable parent and offspring, construction of a sufficiently complete offspring, and evidence that offspring can reproduce in turn. Open-ended evolution additionally requires continuing heritable innovation and diversification beyond a small fixed catalogue; it cannot be established by a few births. Requirements for open-ended evolution include reproductive individuals and viable paths to innovation.

Introducing self-replicating species would add heredity, variation and mutation to the current geometry-only measurements. Selection can arise through differential lineage persistence, while competition for space or explicit resources alters reproductive success. Frequency-dependent interactions mean a type's success depends on its neighbors and abundance. Ecological feedback, coevolution and spatial niche construction can then alter the conditions experienced by descendants. These are proposed mechanisms, not measured findings in this simulator.

Evidence for reproduction would require lineage tracking, causal parent-offspring assignment, repeated independent reproductive cycles, separation from mere fragment emission, and interventions showing what template and environment are necessary. Heritable variation and fitness differences would need explicit measurements. Visually similar structures are candidates for such tests. The present experiments do not implement species, resources, genomes or genealogies and therefore support no ecological or reproduction claim.

12.1 Value Criteria and the Epistemology of Artificial Life

Assessing emergent phenomena requires establishing explicit *value criteria*—normative operational definitions that separate meaningful complexity from trivial dynamics or uncorrelated noise.

Artificial-life research examines constructive models of organization [?]; however, visual resemblance to biological behavior is not an intrinsic property of a lattice, but a projection of observer salience. In cellular automata, dynamical regimes cannot be ranked without embedding subjective or functional preferences directly into the evaluation framework.

The candidate interest metric I formalized in Section 7 follows a specific hierarchy of value criteria. These criteria embody an engineering perspective on complexity, yet they remain extrinsic to the automata themselves. Unlike natural evolutionary systems where value criteria are emergent and self-referential—grounded in reproductive fitness against metabolic constraints—the values applied here are imposed entirely by the external observer. A glider in Conway's Life or a persistent filament in Refractory Life holds no teleological drive or endogenous utility; its "success" is defined solely by the arbitrary weights of our scoring vector.

13 AI usage

An OpenAI Codex agent was used to generate the UI and the majority of the code of the website. The engine was left empty to be implemented in C++. From then on, all coding was done by me. This report is written by me other than details of the web design and generation of figures. All work was checked to ensure accuracy. All generated content is labeled in the report.

14 Limitations

The sample covers only 100 of 262144 binary rules at one size and horizon. Paired trials reduce one source of variability but do not remove RNG limitations or selection bias in the follow-up. All-state and active-only initialization have different expected active densities. Recurrence detection is limited to 16 recent grids. Centroids and bounding boxes have seam artifacts and poor object specificity. Thresholds and interest weights are engineering choices. Noise can sustain externally driven activity. No visual structure detector, lineage tracker, formal proof assistant, infinite-grid analysis was executed as part of the experimental methodology.

15 Future work

Use independent RNG families, more seeds and sizes, preregistered persistence criteria, and finer p spacing where repeated results suggest a crossover. Compare initializers at matched active density as well as matched occupied density. Validate tiny-grid dynamics with an independent model checker. Any reproduction investigation should begin with operational lineage definitions and intervention experiments, not interest-score rankings.

16 Conclusion

The system combines explicit binary and againg state semantics with reproducible paired batch experiments. The exported evidence supports finite, trial-specific comparisons; the report's generated tables and figures quantify those observations. Local correctness checks and reproducible replay

strengthen confidence in implementation behavior while leaving broad claims about emergence, reproduction and evolution open to further investigation.

A Reproduce the Experiment

This is generated with Codex:

From the repository root, run the following commands (use `npm.cmd` on Windows when PowerShell blocks npm scripts):

```
npm test
npm run build
npm run test:integration
npm run experiments
npm run build:site
npm run check:deployment
npm run test:production
python scripts/generate-report.py
cd report
pdflatex main
```

The alternative portable compiler is `tectonic -keep-logs main.tex`. Plot generation requires matplotlib; the local verification renderer uses PyMuPDF. Node, Emscripten and CMake versions are pinned or documented by the existing workflow. Exact run configuration, master and derived seeds, bitmasks, all generation measurements, initial and final grids, categories and thresholds are stored under `results/v2`. Legacy v1 results remain separate historical evidence and use a different noise model; do not combine them with v2.

The C++ suite passed as WebAssembly under Node, including binary compatibility, mask truth tables, Brian's Brain, refractory progression, zero-noise equivalence, seed replay, unique-rule sampling, per-state counts, and full equality after a forced hash collision. The Wasm/worker/HTTP integration suite passed; generated JSON was parsed and all 2880 archived trial schemas, mask encodings and paired sampled grids were validated by the report generator.

The CMake/Emscripten production build and deployment validator passed. Ten headless Edge tests passed across root and repository-subpath HTTP hosting, including downloads, exact replay, all-state initialization, presets and importing saved configurations. The complete LaTeX source compiled with portable Tectonic 0.15.0; page rendering and reference checks are recorded in `report/verification.json`.

Tooling warnings included Playwright's conflicting color-environment flags, Tectonic's missing default Fontconfig configuration and repeated bibliography consistency notices. These did not prevent successful output. Initial sandbox access failures and stale preview processes were resolved by authorized unrestricted tool execution and cleanup. The container-specific artifact marker helper mentioned in the PDF skill was unavailable in this Windows repository; compilation and direct page verification were performed instead. No TODO experiment placeholders remain. Native-machine C++ execution, independent cross-browser accessibility review, and the public GitHub Pages deployment were not performed.